🌀 M003 Extension — The Loop Pattern (Advanced)

Topic: One loop, three counters (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Advanced, after Week 6) · **Time:** about 45 minutes

Until now you placed and filled blocks by choosing **every** coordinate **by hand**. This time **one loop** builds all **7** obsidian walls for you. The loop keeps three counters — **x**, **y**, **z** — and nudges each one by a fixed **step** every pass. That is an **algorithm**: a short list of steps a computer repeats. Load the world, type `SPIRAL` in the chat, and watch 7 walls grow into a pattern.

```
x = 14
y = 0
z = 2
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
    z = z + 2
```

> 🔁 **Big idea:** the fill line **never changes** — it is the same seven times. The walls come out **different** only because **x, y, z change** between passes. Same code, new numbers.

One corner of every wall is **pinned at (14, 0)**. The other corner is **(x, y, z)** — and that is the counter that moves. Each pass the new wall is **2 wider** (x), **1 taller** (y), and **2 deeper** (z) than the last.

> 🔴 Stand on your **home spot** (red block) every run. Move your feet, move the whole pattern.
>
> 🎨 Every wall is **obsidian**.

Trace (or **dry-run**) means: be the computer. Walk the code line by line and write down what each variable holds — **before** you ever run it.

1 · Predict 🔮

Trace the loop. For each iteration write the **START** values (what x, y, z hold when the pass begins — these are the numbers the `fill` line uses) and the **END** values (what they hold after the three update lines run).

Iteration 0's START is given. **The END of one row is the START of the next row** — that is your check.

Iteration	START x	START y	START z	END x	END y	END z
0	14	0	2			
1						
2						
3						
4						
5						
6						

The `fill` line runs *before* x, y, z change. So which x, y, z does iteration 0's wall actually use?

`range(7)` — how many walls does the loop build, and what are the first and last values of `step`?

Which counter makes each wall *taller*? Which makes it *deeper*? Which makes it *wider*? Say why for each.

The wall grows every single pass, yet the `fill` line is never edited. Explain how that is possible.

2 · Spot the Bug 🐛

Each loop below is broken. Trace it in your head, fix it, then explain **why** the original was wrong and **why** your fix works.

Bug A — a counter that never steps. The `z = z + 2` line was deleted. The loop still runs 7 times, but z stays **2** the whole time.

```
x = 14
y = 0
z = 2
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
```

All 7 walls land at z = 2, on top of each other. What do you see in the world instead of a pattern, and why? Write the missing line and where it goes.

Bug B — right steps, wrong order. The three update lines were moved **above** the `fill` line instead of below it.

```
x = 14
y = 0
z = 2
for step in range(7):
    x = x - 2
    y = y + 1
    z = z + 2
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
```

Iteration 0 now fills using x = 12, y = 1, z = 4 — the tiny 1×1 starter step never appears, and the whole pattern shifts. Explain why "update *then* use" and "use *then* update" give different builds.

Bug C *(the tricky one)* — **wrong step size.** It runs with no error, but the 7 separate steps merged into one solid slope with no gaps. A friend changed one line to `z = z + 1`.

```
z = z + 1
```

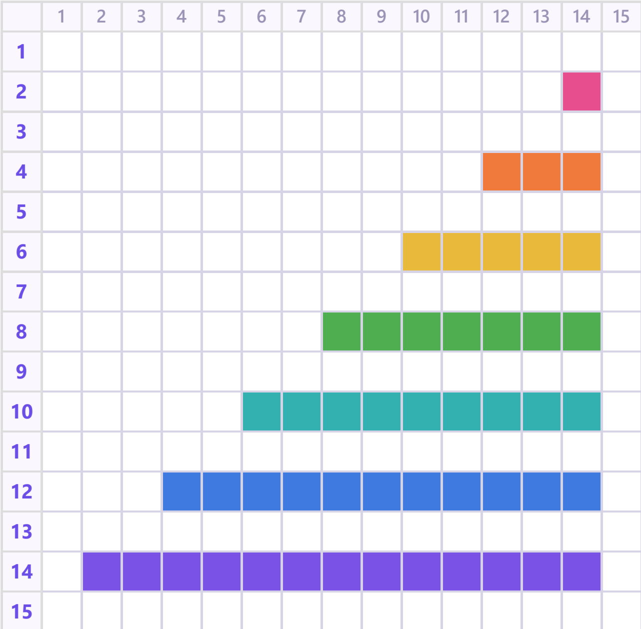
The walls are each 1 block thick. Explain why `z = z + 2` leaves a gap between steps but `z = z + 1` makes them touch. Write the fix.

3 · Show Your Work 📸 🎥

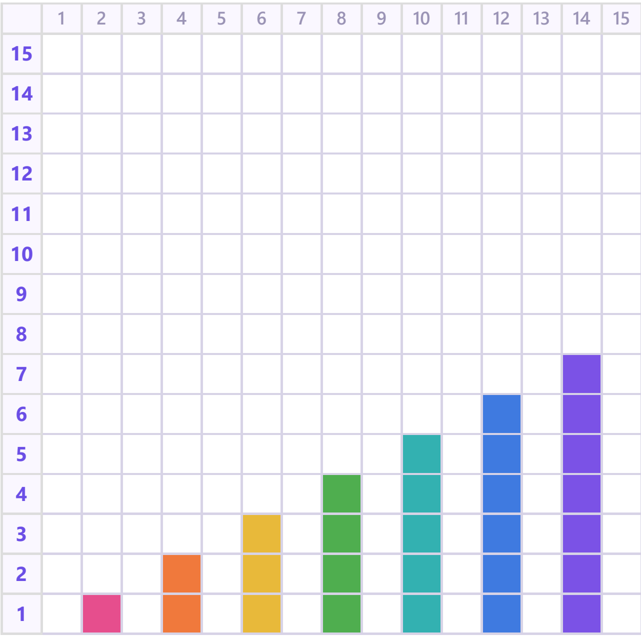
Load the world. Stand on your home spot. Type `SPIRAL` in the chat and watch the loop build all 7 walls. Walk around the pattern — see the tiny 1-block step out front and the tall 7-high wall at the back.

🧱 Same build, two views. Colour = the z value.

👁️ Top view — looking DOWN (x → across, z ↓ deeper)



↔️ Side view (z → deeper, y ↑ up)



colour = z value: 2 4 6 8 10 12 14

Look closely. Reading left to right, the steps climb **1, 2, 3, 4, 5, 6, 7** blocks high — one taller each pass, because `y = y + 1` runs every loop. (Width, the x step, is not drawn here — each step is also wider than the last.)

Modify challenge. Change **one** thing and **predict the new pattern before you run it.** For example: change `range(7)` to `range(10)`, or make the step `y = y + 2` (steeper), or swap `OBSIDIAN` for another block. Write what will move, then build it and check.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what the loop does and what each counter changes.

2 · Your build. Point the camera at the pattern. Name the smallest and biggest step.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your **one** video to teacher on KakaoTalk.